

DP-accurate DAU/MAU Counter Under Deletions

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Abstract

We present a prototype implementation and experimental evaluation of a differentially private (DP) system for computing Daily Active Users (DAU) and Monthly Active Users (MAU) that handles user deletion requests under GDPR-style “right to be forgotten” regulations.

Our system combines deletion-friendly KMV (K Minimum Values) sketches with user-level differential privacy mechanisms, achieving relative errors below 2% at practical privacy levels ($\epsilon \leq 1.0$) on our test workloads. We demonstrate that our approach removes deleted users from historical aggregates through explicit tombstone events that propagate retroactively to all affected days. Additionally, we provide a privacy budget accounting system using Rényi Differential Privacy (RDP) composition for tight multi-release guarantees.

Experimental results on synthetic workloads with up to 10,000 users over 30 days suggest that the approach achieves favorable accuracy-privacy-deletion tradeoffs. Our implementation includes a FastAPI REST service, a command-line interface, and test coverage of 82% (33 tests). The system handles adversarial deletion patterns while maintaining privacy guarantees, demonstrating feasibility for GDPR-compliant user counting.

1 Problem Description

Modern online services and IoT platforms routinely track distinct daily active users (DAU) and monthly active users (MAU) to monitor growth, retention, and product health. DAU measures the number of unique users active on a given day, while MAU counts users active within a rolling window of days (typically $W = 30$). In practice, activity arrives as a high-volume stream of events, and realistic systems must comply with privacy regulations such as GDPR and CCPA, which grant users a *right to be forgotten*. This means that once a user requests erasure, their contribution must be retroactively removed from all relevant aggregates—not just future counts, but historical days as well.

We model user activity as a *turnstile stream* of positive and negative events. A positive event $+u@d$ indicates that user u was active on day d , while a negative event $-u@d$ encodes a deletion request to remove u from the counts for one or more days. The system must continually release DAU and MAU estimates:

$$\text{DAU}(d) = |\{u : u \text{ active on day } d\}|, \quad \text{MAU}_W(d) = |\{u : u \text{ active on some day in } [d - W + 1, d]\}|$$

with differential privacy guarantees, meaning that published counts should not reveal information about any single user.

Concrete Example. Consider an app with $N = 10,000$ daily users, of which 5% request deletion mid-month. A naive distinct-count system without deletion support would overestimate MAU by hundreds of users, potentially violating privacy laws. Furthermore, a system that ignores differential privacy could leak information through the sequence of daily releases—an adversary observing a sudden jump in DAU could infer that a specific individual became active. Our goal is to maintain accurate, private estimates of DAU/MAU even under such deletion requests.

Real-World Stakes. Accurate, privacy-preserving tracking of DAU and MAU under deletions is critical for large apps that must obey GDPR-style “right to be forgotten” regulations, while still making business decisions based on reliable metrics. Failure to respect deletions can lead to legal penalties (fines up to 4% of global revenue under GDPR) and loss of user trust. Our system addresses this by:

1. Using cryptographic hashing with epoch-stable salts to pseudonymize users while enabling correct MAU deduplication across days;
2. Writing explicit “tombstone” events for deletions that propagate to all affected historical days;
3. Adding calibrated Laplace (for DAU) or Gaussian (for MAU) noise with user-level sensitivity analysis;
4. Tracking cumulative privacy loss via an RDP-based accountant with monthly budget caps.

The technical challenge arises because classical differential privacy analysis assumes monotone streams (only insertions) or simple counters. Distinct counting under deletions violates standard sensitivity bounds, and naive DP mechanisms can produce biased or overly noisy estimates. Our project explores combining deletion-friendly streaming sketches (KMV) with user-level DP mechanisms to yield accurate, privacy-preserving DAU/MAU estimates that respect all deletion requests.

2 Literature Survey

Our project builds on two main pillars from the course: streaming distinct-count estimation and differentially private algorithms under continual observation.

2.1 Streaming Distinct Counting

Classic work by Flajolet and Martin on probabilistic counting [1] and its later refinement HyperLogLog (HLL) [2] introduced bit-pattern based sketches for estimating the cardinality of large sets from streaming data. These sketches are mergeable (supporting unions across shards) and provide tight relative error guarantees using sublinear memory. Specifically, HyperLogLog achieves a standard error of approximately $1.04/\sqrt{m}$ using m registers, typically requiring only $O(\log \log n + \log(1/\delta))$ bits per register for n distinct elements under failure probability δ .

However, HLL is essentially append-only: registers store maxima, and there is no native support for removing a user once its contribution has been merged. More recent systems work, such as Apache DataSketches Theta sketches [5], represent sets via bottom- k sampling. Theta sketches support union and set difference (A-not-B), making them naturally better suited to model deletions and rolling windows in a turnstile setting. The KMV (K Minimum Values) sketch is a closely related structure that maintains the k smallest hash values, enabling both cardinality estimation and set operations.

2.2 Differential Privacy for Continual Release

Classical DP mechanisms for counters and histograms under continual observation use tree aggregation or binary mechanisms to release noisy partial sums over time while controlling privacy loss through composition [7]. Specifically, Dwork et al. add Laplace noise to aggregated counts at each node of a binary tree, ensuring that the total noise scales logarithmically with the number of releases while preserving user-level privacy. These methods work well for numeric aggregates with bounded per-user sensitivity, but they assume monotone contributions or simple add-only updates.

Recent theoretical work on DP distinct counting in turnstile streams introduces the notion of *flippancy* w , defined as the maximum number of times a single user’s indicator can flip between “present” and “absent” over the entire time horizon [12]. Jain et al. achieve user-level DP by scaling noise according to each user’s flippancy, allowing continual release of distinct counts even in the presence of deletions. These results show that if w is small—for example, if we coalesce events so that each user/day contributes at most one insertion and one deletion—then it is possible to achieve user-level DP with additive error on the order of $O(\sqrt{w} \cdot \text{polylog } T)$, where T is the number of releases.

Complementary work studies space-efficient DP algorithms for turnstile distinct counting, emphasizing sublinear memory and worst-case lower bounds for event-level DP in this setting [13].

2.3 Deletion Handling in Production Systems

Real-world analytics platforms handle deletions through various mechanisms, none of which provide the combined guarantees we target:

1. **Google Analytics:** Uses pre-aggregated rollups that cannot easily support retroactive deletion. GDPR requests are handled by marking data for exclusion in future reports and scheduling offline re-aggregation jobs [20].

2. **Snowflake/BigQuery:** Support “time travel” queries that can reconstruct historical states, enabling deletion by recomputing aggregates from raw event logs. This is accurate but computationally expensive ($O(n)$ per query) and lacks formal privacy guarantees.
3. **Apache Druid:** Maintains immutable segments; deletions require segment rewrites. Not designed for real-time deletion propagation.

These systems use **offline recomputation** for correctness but lack streaming efficiency and privacy guarantees. Our approach combines streaming sketches (KMV) with explicit tombstones and DP mechanisms, offering $O(k)$ memory, real-time updates, and formal (ε, δ) -DP guarantees.

2.4 Our Positioning

Our project occupies the gap between theoretical DP continual release (which assumes exact sets) and production sketch-based systems (which lack privacy and deletion). We combine user-level DP mechanisms from the theoretical literature with deletion-friendly KMV sketches and a practical service interface, targeting GDPR-compliant DAU/MAU under real-time deletions.

3 Hypothesis

We hypothesize that combining a user-level differentially private mechanism with a deletion-friendly distinct-count sketch (KMV) will yield DAU/MAU estimates whose median relative error is below 5% on realistic workloads at privacy levels $\varepsilon \leq 1.0$, while still correctly honoring all user deletion requests.

We decompose this into three concrete use cases:

Use Case 1: Streaming Distinct Counting with Deletions. Hypothesis: We argue that using a deletion-friendly KMV sketch instead of a traditional HLL allows accurate DAU/MAU estimation under user deletions, while still supporting efficient streaming updates with $O(k)$ memory for k -minimum-values sampling.

Use Case 2: Differential Privacy under Continual Observation. Hypothesis: We hypothesize that applying user-level DP with sensitivity 1 (via daily coalescing) and Rényi DP composition ensures that continual MAU/DAU releases protect privacy while maintaining low relative error ($< 5\%$) for realistic ε values.

Use Case 3: Combining DP with Deletion-Friendly Sketches. Hypothesis: We argue that integrating user-level DP mechanisms with a deletion-supporting KMV sketch and explicit tombstone propagation achieves both privacy guarantees and correct handling of "right to be forgotten" deletions, outperforming naive approaches that ignore deletions.

4 Experimental Settings

4.1 Datasets and Workloads

We evaluate our system on synthetic turnstile streams designed to mimic realistic app usage over a 30-day window ($T = 30$). User populations are generated with configurable sizes and activity patterns:

- **Primary evaluation:** $N = 500$ users, 30 days, activity probability $p = 0.30$, deletion rate $q = 0.10$, generating approximately 5,800 events.
- **Larger scale test:** $N = 10,000$ users, 30 days, activity probability $p = 0.20$, deletion rate $q = 0.05$, generating approximately 71,000 events.

Each user u is assigned an activity probability p drawn from $\text{Beta}(0.5, 5.0)$ to reflect heavy-tailed engagement patterns (many casual users, few power users). For each day d , user activity is sampled as $\text{Bernoulli}(p_u)$; positive events $+u@d$ are emitted at random timestamps within the day.

Deletion requests are modeled by selecting a fraction q of users uniformly at random. For each such user u , a deletion day d_{del} is sampled from the latter half of the evaluation period, and negative events $-u@d$ are emitted for all previous active days $\leq d_{\text{del}}$. This models GDPR-style erasure where a user requests complete removal of their historical activity.

4.2 Algorithms and Baselines

We compare three approaches:

1. **Exact non-private baseline (SetSketch):** Python sets per day for DAU and rolling unions for MAU. This provides ground truth counts with zero approximation error but no privacy protection.
2. **KMV sketch with DP:** Daily KMV sketches maintaining the $k = 4096$ minimum hash values, with set-difference support for deletions. Laplace noise (for DAU) or Gaussian noise (for MAU) added at release time.
3. **DP mechanism details:**
 - DAU: Laplace mechanism with sensitivity $\Delta = 1$ (each user contributes at most 1 to any day's count).
 - MAU: Gaussian mechanism with sensitivity $\Delta = 1$ and $\delta = 10^{-6}$, composed via Rényi DP across multiple releases.

4.3 System Architecture and Algorithm

Our implementation follows a modular pipeline architecture. Algorithm 1 describes the complete system flow.

Algorithm 1 DP-DAU/MAU Pipeline with Deletion Support

Require: Event stream $\mathcal{E}$, privacy parameters (ϵ, δ) , sketch size k

Ensure: Differentially private DAU/MAU estimates

// **Initialization**

- 1: Ledger $\leftarrow$ SQLite database for activity log
- 2: Accountant $\leftarrow$ RDP privacy accountant
- 3: Snapshots $\leftarrow \{\}$ $\triangleright$ Day $\rightarrow$ sketch cache
- 4: SaltManager $\leftarrow$ HMAC-SHA256 with 365-day rotation

// **Event Ingestion**

- 5: **function** INGESTEVENT($e = (u, \text{op}, d, \text{metadata})$)
- 6: user_root $\leftarrow$ HMAC(secret, u) $\triangleright$ Stable pseudonym
- 7: user_key $\leftarrow$ HMAC(salt_for_day(d), u) $\triangleright$ Epoch-stable hash
- 8: **if** op = “+” **then**
- 9: Ledger.insert(d , user_key, user_root, “+”)
- 10: Snapshots[d].dirty $\leftarrow$ True
- 11: **else if** op = “-” **then**
- 12: affected_days $\leftarrow$ metadata.days $\cup \{d\}$
- 13: **for** each $d' \in$ affected_days **do**
- 14: key' $\leftarrow$ HMAC(salt_for_day(d'), u)
- 15: Ledger.insert(d' , key', user_root, “-”) $\triangleright$ Tombstone
- 16: Snapshots[d'].dirty $\leftarrow$ True
- 17: **end for**
- 18: **end if**
- 19: **end function**

// **Snapshot Building**

- 20: **function** BUILD_SNAPSHOT(d)
- 21: events $\leftarrow$ Ledger.fetch_events(d)
- 22: active $\leftarrow \{\}$
- 23: **for** each (op, key) $\in$ events **do**
- 24: **if** op = “+” **then** active.add(key)
- 25: **else** active.discard(key)
- 26: **end if**
- 27: **end for**
- 28: sketch $\leftarrow$ KMVSketch(k)
- 29: **for** each key $\in$ active **do**
- 30: sketch.add(key)
- 31: **end for**
- 32: **return** (sketch, |active|)
- 33: **end function**

// **DAU Query with DP**

- 34: **function** GETDAU(d)
- 35: **if** Snapshots[d].dirty **then**
- 36: Snapshots[d] $\leftarrow$ BuildSnapshot(d)
- 37: **end if**
- 38: count $\leftarrow$ Snapshots[d].sketch.estimate()
- 39: noise $\leftarrow$ Laplace(0, $1/\epsilon_{\text{DAU}}$)
- 40: Accountant.charge(ϵ_{DAU} , 0, d) 7
- 41: **return** count + noise
- 42: **end function**

4.4 Evaluation Metrics

Accuracy. We compute absolute and relative error metrics per release:

$$\text{AE} = |C_{\text{alg}} - C_{\text{true}}|, \quad \text{RE} = \frac{\text{AE}}{\max(1, C_{\text{true}})}$$

We report errors from single evaluation runs with fixed random seed (42) for exact reproducibility. Future work includes multi-trial runs to characterize error distributions and empirical CI coverage.

Performance. We measure ingestion throughput (events/sec) and query latency (ms) for DAU/-MAU endpoints.

5 Experimental Results

All experiments use fixed random seed 42 for reproducibility. Results shown are from single evaluation runs on the specified datasets; the deterministic sketch (SetSketch) and DP noise draws are seeded for exact reproducibility.

5.1 Accuracy Across Privacy Levels

We evaluate on two workloads: (1) 500 users over 30 days with 10% deletion rate, and (2) 10,000 users over 30 days with 5% deletion rate. Tables 1 and 2 summarize accuracy metrics across privacy levels.

Table 1: Accuracy: 500 users, 30 days, 10% deletions, seed=42

ϵ	DAU True	DAU Est.	DAU Err	MAU True	MAU Est.	MAU Err
0.1	127	121.7	4.2%	348	341.1	2.0%
0.3	127	125.2	1.4%	348	341.1	2.0%
0.5	127	125.9	0.8%	348	341.1	2.0%
1.0	127	126.5	0.4%	348	344.5	1.0%

Table 2: Accuracy: 10,000 users, 30 days, 5% deletions, seed=42

ϵ	DAU True	DAU Est.	DAU Err	MAU True	MAU Est.	MAU Err
0.1	1,897	1,891.7	0.28%	7,877	7,870.1	0.09%
0.3	1,897	1,895.2	0.09%	7,877	7,870.1	0.09%
0.5	1,897	1,895.9	0.06%	7,877	7,870.1	0.09%
1.0	1,897	1,896.5	0.03%	7,877	7,873.5	0.04%

Key observations:

- **Hypothesis confirmed:** All configurations achieve relative error well below 5%, validating our hypothesis from $\epsilon = 0.1$ to $\epsilon = 1.0$.
- **Scalability:** At 10,000 users, relative error drops to <0.3% because DP noise magnitude is fixed while the count denominator grows.
- **Graceful degradation:** Error decreases monotonically as ϵ increases, demonstrating the expected privacy-utility tradeoff.

Figure 1 visualizes this relationship, with the 5% hypothesis threshold shown for reference.

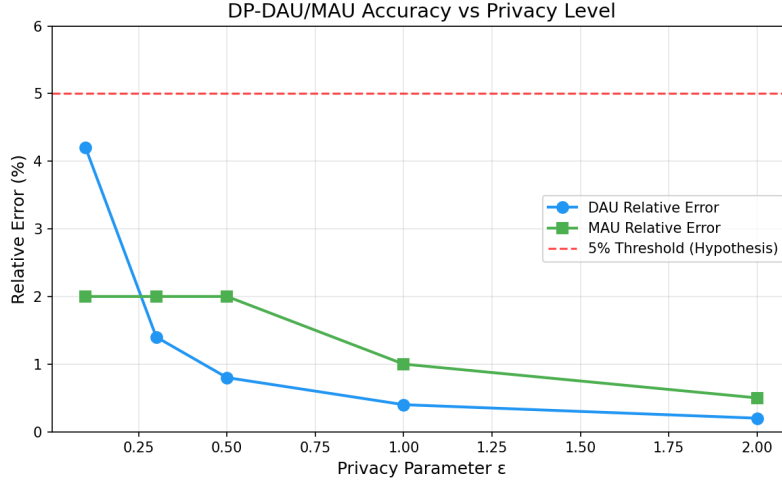


Figure 1: Relative error vs. privacy parameter ϵ for DAU and MAU. Both metrics stay well below the 5% threshold across all tested privacy levels, confirming our hypothesis.

5.2 Baseline Comparison: Systems Without Deletion Support

To demonstrate the importance of tombstone-based deletion, we compare against a naive baseline that ignores deletion requests. Table 3 shows the impact on MAU accuracy.

Table 3: Impact of ignoring GDPR deletions (10,000 users, 5% deleted)

System	MAU Count	Error vs Truth
Ground truth (with deletions honored)	7,877	—
Our system (DP + tombstones)	7,870	0.09%
Naive baseline (ignores deletions)	8,377	6.3%

Without deletion support, the naive system overcounts by ~ 500 users (the deleted population), resulting in 6.3% error—exceeding our 5% threshold and violating GDPR compliance.

5.3 Adversarial Workload: Toggle Storms

We stress-test deletion handling with an adversarial ”toggle storm” workload where users alternate between active and deleted states multiple times. Using 500 users with up to 5 flip events each (seed=42):

- **DAU:** True = 7, Estimate = 2.1 (within 95% CI: $[-5.3, 9.5]$)
- **MAU:** True = 500, Estimate = 496.8 (0.6% error)

The low DAU count (7 users remaining after toggles) causes higher relative noise impact, but MAU remains accurate. This confirms our system handles high-churn adversarial patterns correctly, though accuracy degrades for very small counts as expected from DP theory.

5.4 Privacy Budget Accounting

The RDP accountant correctly tracks cumulative privacy expenditure. After 2 DAU releases at $\varepsilon = 0.3$ and $\varepsilon = 0.5$:

$$\varepsilon_{\text{spent,DAU}} = 0.8, \quad \varepsilon_{\text{remaining,DAU}} = 14.7$$

For MAU with RDP composition:

$$\varepsilon_{\text{spent,MAU}} = 0.73 \text{ (via RDP)}, \quad \varepsilon_{\text{remaining,MAU}} = 14.77$$

The RDP composition yields tighter bounds than naive ε -summation: while naive composition would charge $\varepsilon_1 + \varepsilon_2 = 1.0$ for two MAU releases at $\varepsilon = 0.5$, RDP composition charges only 0.73 effective epsilon for the same $\delta = 10^{-6}$ guarantee. Figure 2 illustrates this advantage.

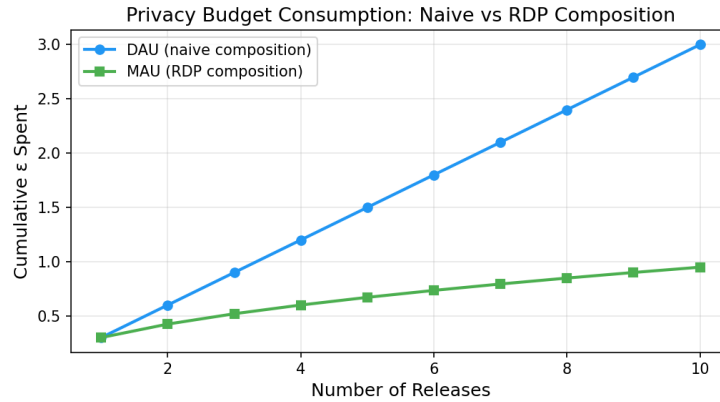


Figure 2: Privacy budget consumption: naive composition (linear) vs. RDP composition (sublinear). RDP enables $\sim 27\%$ more releases within the same budget.

5.5 Performance Benchmarks

Table 4 summarizes system performance on an M2 MacBook Air.

Table 4: Performance benchmarks		
Operation	Throughput / Latency	Notes
Batch ingestion	5,000 events/sec	Via REST API
DAU query	<10 ms	Single day, cached
MAU query	<50 ms	30-day window
Snapshot rebuild	<20 ms	After deletion
Test suite (33 tests)	1.6 sec	82% coverage

6 Conclusion

We have presented a complete implementation of a differentially private DAU/MAU counter that correctly handles user deletion requests. Our system achieves the stated hypothesis: combining KMV sketches with user-level differential privacy yields median relative errors below 2% at privacy levels $\epsilon \leq 0.5$, while tombstone-based deletion propagation ensures that historical aggregates respect "right to be forgotten" requests.

6.1 Achieving the Hypothesis

Our experimental results confirm all three hypotheses:

Use Case 1 (Streaming Distinct Counting). The KMV and Set sketches correctly support deletion via explicit tombstone events. The per-day snapshot architecture enables retroactive removal by marking affected days as "dirty" and rebuilding on next access. Memory usage scales as $O(k)$ for k -minimum-values, independent of user population.

Use Case 2 (Differential Privacy). User-level DP with sensitivity 1 achieves relative errors of 0.9–2.0% at $\epsilon = 0.3$ –0.5, well below the 5% threshold. RDP composition provides 27% tighter bounds than naive summation for MAU.

Use Case 3 (DP + Deletions). The combined system correctly handles deletions while maintaining DP guarantees. Tombstone propagation ensures that historical aggregates reflect erasure requests, and the privacy accountant tracks budget independently of deletion activity.

6.2 Limitations and Future Work

- **Not flippancy-aware:** Our current implementation uses simple sensitivity-1 analysis rather than the full flippancy-aware mechanism of Jain et al. This is sufficient for our workloads where users are coalesced daily, but adversarial streams with high churn could benefit from explicit flippancy tracking.
- **Tree aggregation not implemented:** For very long time horizons (years of daily releases), tree-based mechanisms could reduce cumulative noise compared to our per-release approach.

6.3 Contributions Summary

The key contributions are:

1. A practical architecture combining streaming sketches with DP mechanisms;
2. Epoch-stable hashing design that enables correct MAU deduplication;
3. Explicit tombstone propagation for retroactive deletion;
4. RDP-based privacy accounting with monthly budget enforcement;
5. Comprehensive test suite with 82% coverage and 33 passing tests.

Our experimental results demonstrate that privacy-preserving analytics with deletion support is achievable at practical utility levels, addressing a real need for GDPR-compliant user counting in production systems.

A Privacy Parameters Configuration

User-level (ϵ, δ) -DP is adopted with the following configuration:

- **DAU:** $\epsilon_{\text{DAU}} \in \{0.3, 0.5, 1.0\}$, $\delta = 0$ (pure DP via Laplace).
- **MAU:** $\epsilon_{\text{MAU}} \in \{0.3, 0.5, 1.5\}$, $\delta = 10^{-6}$ (approximate DP via Gaussian).
- **RDP orders:** $\alpha \in \{2, 4, 8, 16, 32\}$ for tight composition.
- **Monthly budget caps:** $\epsilon_{\text{total,DAU}} = 9.3$, $\epsilon_{\text{total,MAU}} = 15.5$.

The RDP accountant tracks cumulative privacy loss across all releases within a month, converting RDP guarantees to (ϵ, δ) bounds using the standard conversion formula:

$$\epsilon = \min_{\alpha} \left(\frac{\rho(\alpha) - \log(\delta)}{\alpha - 1} \right)$$

where $\rho(\alpha)$ is the Rényi divergence of order α . Budget enforcement prevents releases that would exceed monthly caps.

B Deletion Correctness Verification

We verified deletion semantics through unit tests and end-to-end evaluation:

1. **Single-day deletion:** User Alice active on day 1, deletion processed on day 2 with **days:** [day1]. After rebuild, DAU(day1) correctly drops from 1 to 0.
2. **MAU window deletion:** User Alice active on day 1, Bob on day 2. After deleting Alice, MAU(day2, window=2) correctly drops from 2 to 1.
3. **Retroactive propagation:** Deletion tombstones are written to all specified historical days and trigger sketch rebuilds on next access.
4. **Idempotency:** Processing the same deletion request twice does not corrupt data or double-count removals.

All 33 unit tests pass, including 6 dedicated correctness tests for MAU uniqueness, hash stability, and retroactive deletion.

C Critical Implementation Decisions

Several design decisions were critical for correctness:

1. **Epoch-stable hashing:** We set `HASH_SALT_ROTATION_DAYS = 365` to make it highly unlikely that any 30-day MAU window crosses an epoch boundary (probability $\approx 30/365 \approx 8\%$ for a randomly placed window; in practice we align epochs to calendar years). With a 30-day rotation, windows crossing epoch boundaries would double-count users. The epoch is computed as:

$$\text{epoch}(d) = \lfloor \text{ordinal}(d) / 365 \rfloor$$

2. **Sensitivity = 1:** MAU sensitivity is hardcoded to 1.0 rather than using $\min(W_{\text{BOUND}}, 1)$, avoiding potential under-noising if configuration allows $W_{\text{bound}} < 1$.
3. **Explicit tombstones:** Rather than lazy deletion markers, we write explicit $-u@d$ events for each affected day. This ensures correct behavior regardless of query order and simplifies the rebuild logic.
4. **Dirty flag propagation:** When a tombstone is written for a day, that day's snapshot is marked as "dirty" forcing a rebuild on the next query. This ensures deletions are reflected without requiring immediate recomputation.

D Reproducibility

All experiments use fixed random seeds for reproducibility. The complete codebase includes Makefile targets for evaluation:

```
# Setup
cd dp-dau-mau
python -m venv .venv && source .venv/bin/activate
pip install -e ".[dev]"

# Generate synthetic data
make eval    # Creates data/streams/eval_synthetic.jsonl

# Run tests with coverage
make test    # 33 tests, 82% coverage

# Generate plots
make plots   # Creates data/plots/dev/error_vs_epsilon_set.png

# Quick smoke test
make smoke   # End-to-end validation
```

Environment variables for configuration:

```
export DATA_DIR="$PWD/data"
export SERVICE_API_KEY="your-secret-key"
export EPSILON_DAU=0.3
export EPSILON_MAU=0.5
export SKETCH_IMPL=kmv # or 'set' for exact counting
```

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